

Traditional Atomistic Simulation: The Challenge



Extremely time-consuming

Replaying vast numbers of low-impact micro-events before reaching decisive states.



Requires massive resources

Typically needs supercomputers with thousands of CPU/GPU cores and long wall-clock time.



Compute wasted on uninformative events

Budget is consumed by local fluctuations that do not change the system's structural outcome.

Compute is trapped in meaningless event replay instead of advancing along the true evolutionary backbone.



Days to Weeks

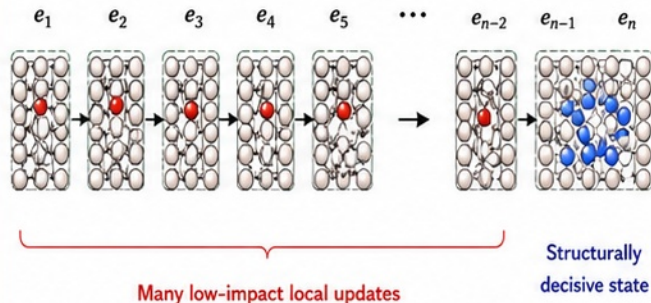


Supercomputer

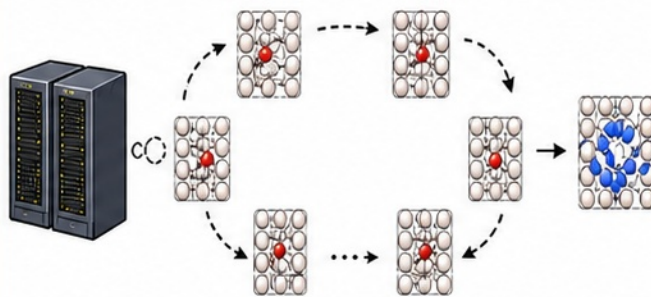
Traditional Event-by-Event Simulation

1 Event-by-event (micro-resolution) advancement

The simulator advances through every micro-event sequentially.



2 Simulation budget is consumed at micro-event resolution before reaching structurally decisive states.



Host atom



Vacancy / site



Mobile atom / defect



Cluster / product

AtomWorld-Mirror: Macro-step World Modeling

1 Learn from micro-evolution and predict Multi-K macro-steps

From short micro-evolution segments, the world model predicts key events K steps ahead (multi-K macro-step), including structure edits and time.

Observe short micro-evolution
(teacher / data)

Predict Multi-K macro-steps
(student / inference)



Physically Constrained



Local reachability



Inventory conservation

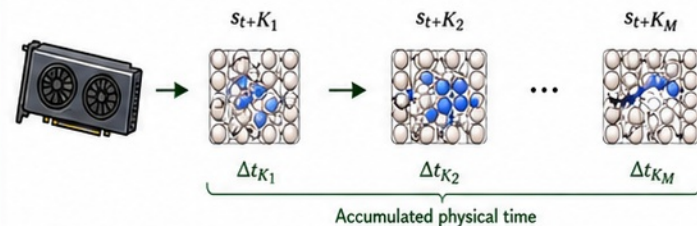


Continuous-time consistency

Outputs per macro-step

- ✓ Key structural edits (sparse)
- ✓ Accumulated physical time Δt
- ✓ Physically reachable next state

2 Budgets are spent on reaching decisive states, not on replaying micro-events.



1 × A100 GPU



Hours
(not days or weeks)